

HR Workforce Analytics Report

Workforce composition, employee status, compensation, hiring history, and selected employee measures



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Sources: Human Resources Data Set, Kaggle (Rhuebner), 2020: <https://www.kaggle.com/datasets/rhuebner/human-resources-data-set>

How I approached the workforce analysis

I used an exploratory process to understand the workforce, identify meaningful patterns, and define the questions HR should investigate next.

01

Profile the workforce

Review headcount, department, gender, employee status, and salary.

02

Compare employee groups

Compare active and former records, departments, and salary distributions.

03

Screen relationships

Use scatter plots and correlations to identify patterns worth investigating.

04

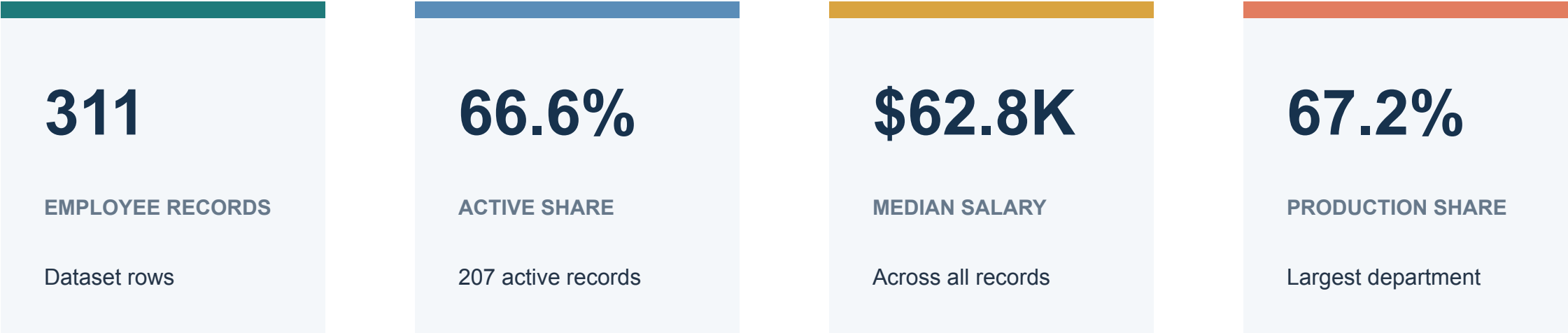
Interpret with context

Separate observation from causation, note limitations, and define next questions.

Goal: understand the workforce, identify meaningful patterns, and determine what HR should investigate next.

The dataset provides a workforce data

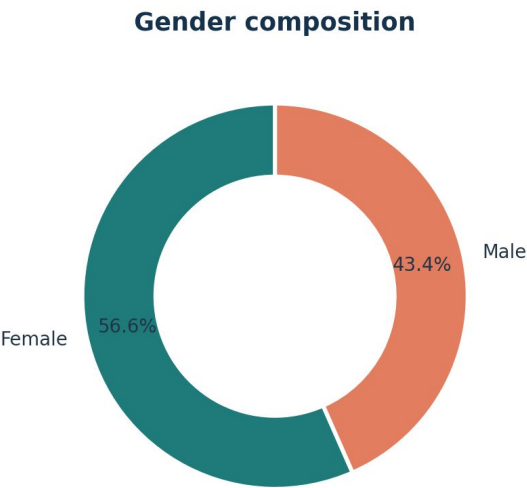
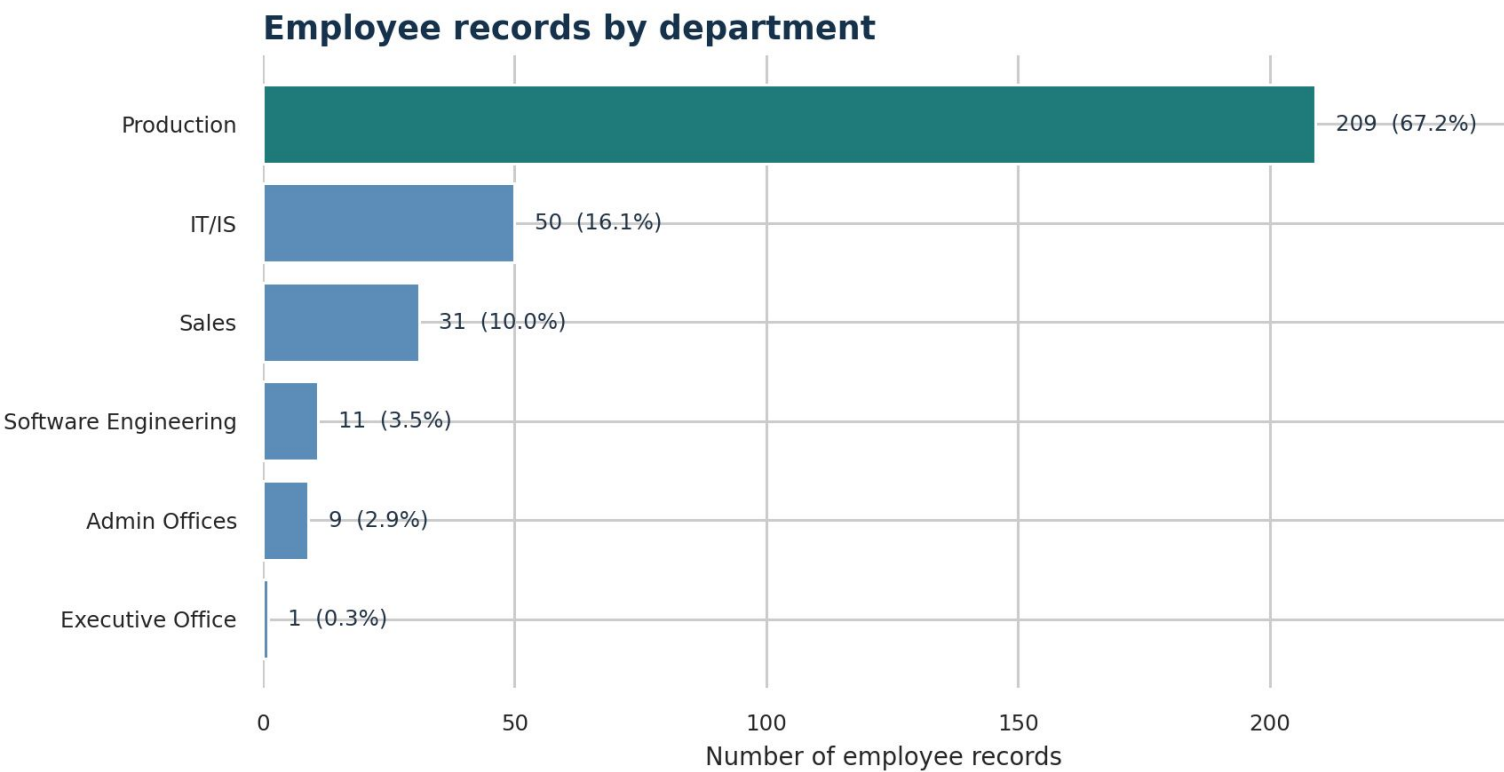
This analysis identifies patterns and frames follow-up questions for HR. Observed relationships are treated as associations, not proof of cause or prediction.



Important interpretation

The 33.4% former-employee share is a historical record mix, not an annual turnover rate. A true rate needs a defined period and average headcount.

Production accounts for two-thirds of employee records



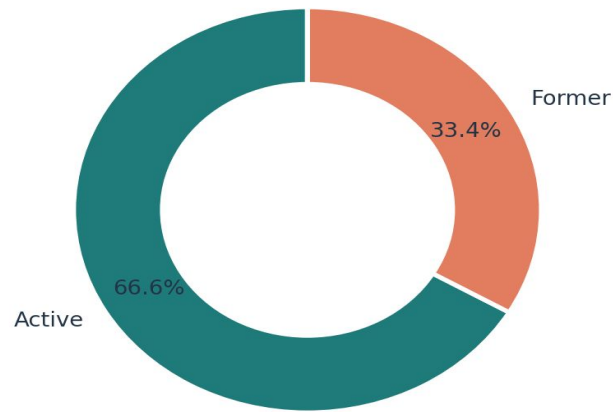
What it means

Production has most of the employees. HR should also review each department separately

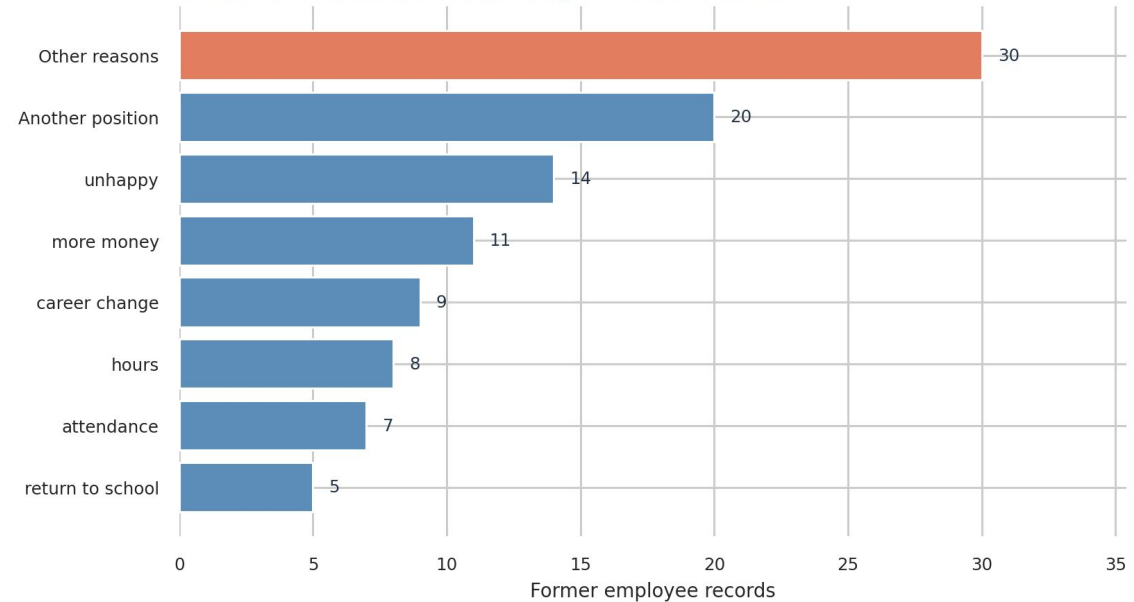
Aggregate gender balance does not establish equity within departments, levels, pay, or outcomes.

Former employees represent 33.4% of the records

Record status in the dataset



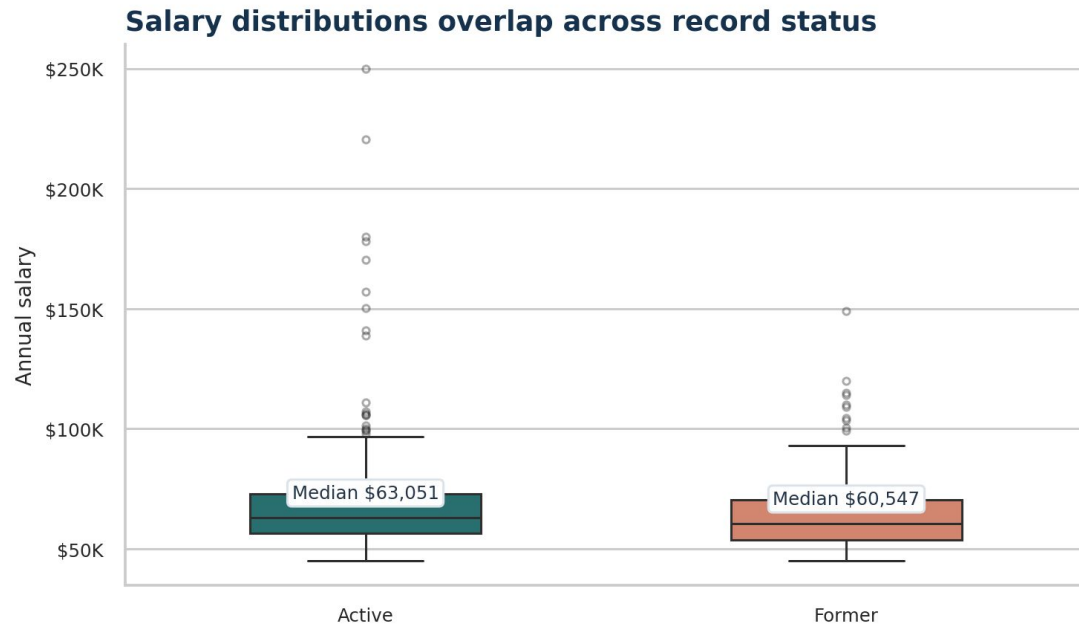
Most common recorded separation reasons



- The dataset includes 207 active employee records, representing 66.6% of the dataset, and 104 former employee records, representing 33.4%
- Top recorded reasons: another position (20), unhappy (14), and more money (11).
- These three reasons represent 43.3% of former-employee records.

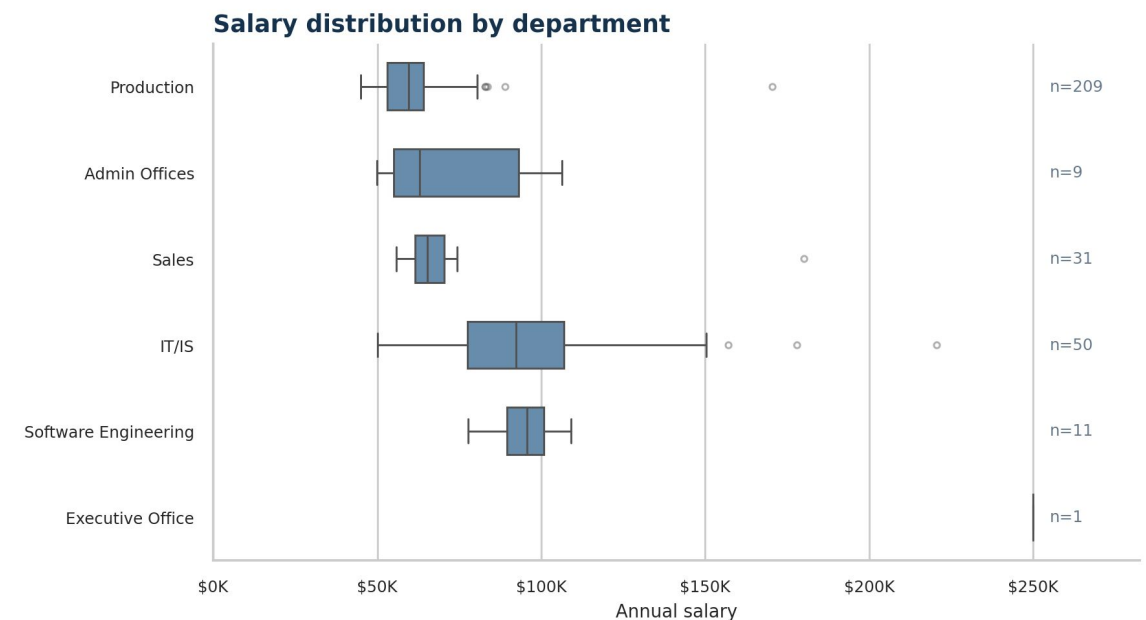
Since the dataset does not have a clear reporting period, we cannot treat the former-employee percentage as a turnover rate. For further analysis, we could calculate the actual turnover rates for a specific period and compare voluntary and involuntary turnover across departments, tenure groups, managers, and job families.

Salary patterns depend on comparable jobs and departments



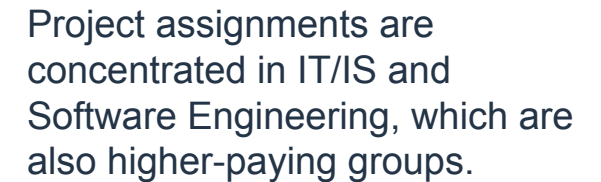
Salary alone does not clearly separate active and former records

Median salary: \$63,051 for active employees vs. \$60,547 for former employees, while the overall salary distributions remain similar.



Salary differences across departments need more context

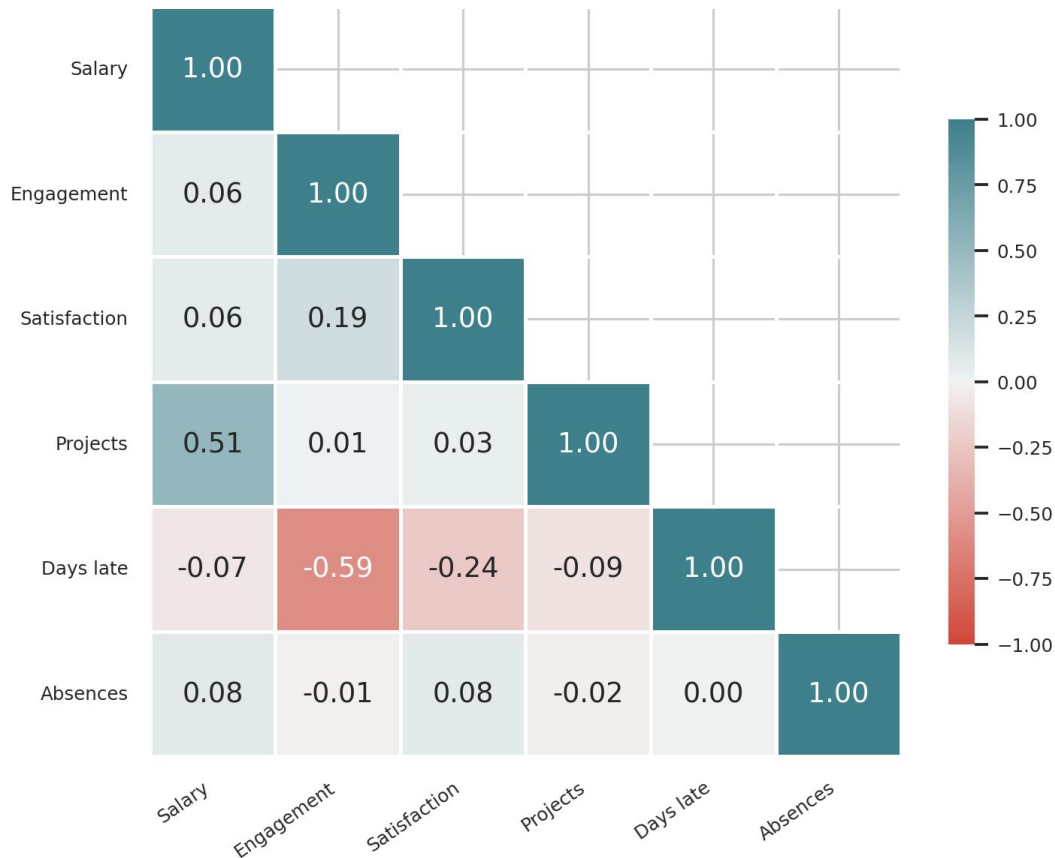
Departments include different jobs and have very different group sizes. For example, the Executive Office has only one employee record. A fair comparison should focus on similar roles and job levels.



Compare similar roles within the same department to better understand the relationship between projects and salary.

Heatmap focuses on meaningful numeric measures

Exploratory correlations among meaningful numeric measures



Two relationships stand out

-0.59

Engagement and
days late

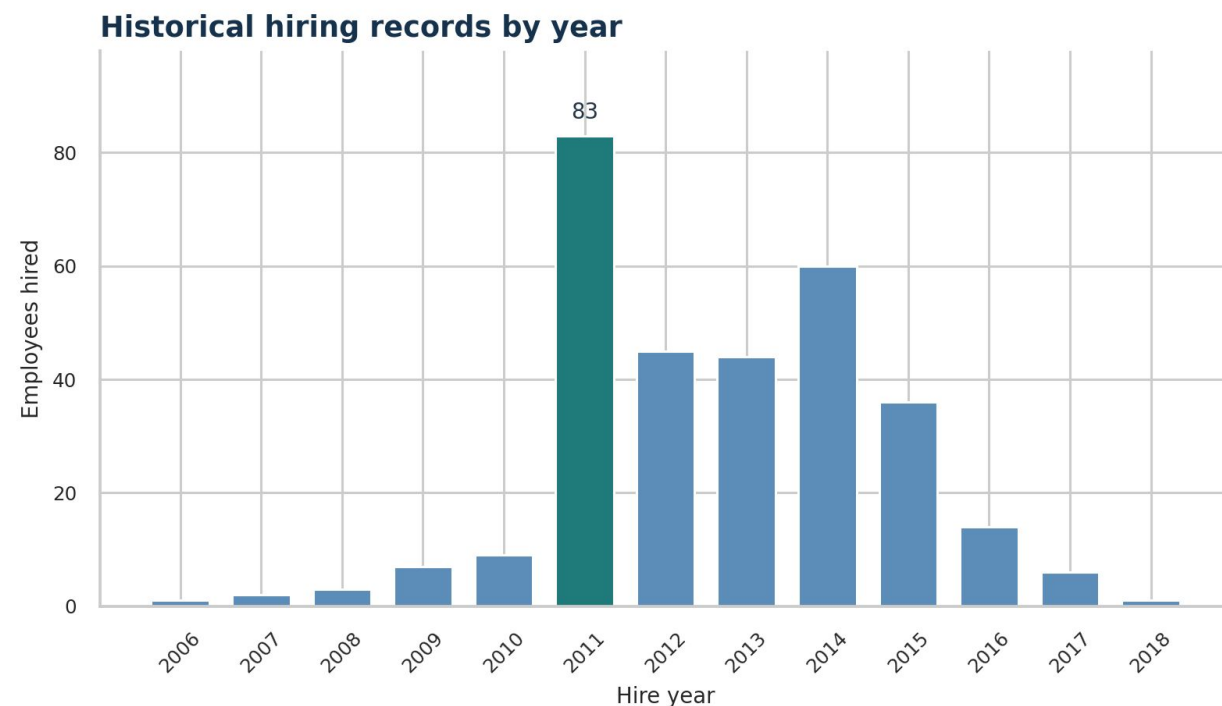
0.51

Salary and special
projects

- Category codes such as DepartmentID, PositionID, and ManagerID are excluded.
- Correlation identifies associations; it does not prove causation or predictive value.

The heatmap highlights possible relationships for further analysis, rather than providing final conclusions.

Validated data turns a workforce snapshot into useful analysis



The dataset shows 83 hires in 2011 and 60 in 2014, followed by fewer recorded hires through 2018. Confirm the reporting cutoff and business context before labeling the decline a workforce risk.

Analysis roadmap

- 1 Validate definitions, dates, owners, and reporting cutoff.
- 2 Build period-based turnover rates with average headcount.
- 3 Segment voluntary exits by department, tenure, job family, and manager.
- 4 Review compensation within comparable roles and levels.
- 5 Connect hiring, exits, vacancies, and workforce plans.

HR analytics helps us ask better questions

Overall, this analysis helped me understand the workforce and identify areas that HR may want to investigate further. My main takeaway is that HR analytics is not just about finding patterns. It is also about understanding the limitations of the data and knowing what additional information is needed before making a decision.